

Xitong Sun

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Summary

M.S.E. student in Mechanical Engineering at UT Austin with research experience and interests in **piezoelectric power conversion** and **high-performance power converter modeling and control**.

Education

The University of Texas at Austin (UT)

Austin, TX

M.S.E. in Mechanical Engineering

Aug. 2025 – May 2027 (expected)

- **GPA:** 3.93/4.0
- **Advisor:** Prof. **Ruochen Lu** [Link]
- **Relevant Courses:** Power Electronic Devices and Systems; Power Management Integrated Circuits (PMIC); ULSI Technology; Microelectromechanical Systems (MEMS).
- **Teaching Experience:** Teaching Assistant for ECE 402 Introduction to Electrical Engineering; ME 302 Introduction to Engineering Design and Graphics.

Southern University of Science and Technology (SUSTech)

Shenzhen, China

B.E. with Honors in Microelectronics

Sep. 2021 – Jul. 2025

- **GPA:** 3.76/4.0
- **Advisor:** Prof. **Fei Wang** [Link]

University of California, Berkeley

Berkeley, CA

Summer Extension Program

Jun. 2023 – Aug. 2023

- **GPA:** 4.0/4.0

Selected Publications

- [1] **X. Sun**, J. Li, T. Cheng and F. Wang*, "MEMS Gas Sensor Arrays with Multilayer of Nanomaterials Patterned by Wafer-Level Photolithography Process," *2025 23rd International Conference on Solid-State Sensors, Actuators and Microsystems (Transducers)*, Orlando, FL, USA, 2025, pp. 1061–1064. [Link]
- [2] J. Li[†], **X. Sun**[†], T. Cheng and F. Wang*, "MEMS Gas Sensor with On-chip Electrospun Ru-SnO₂ Nanospheres Patterned by Photolithography," *2024 IEEE SENSORS*, Kobe, Japan, 2024, pp. 1–4. [Link]

Honors

- **Britton R. Birmingham Graduate Fellowship in Engineering, UT** *2025, 2026*
- Outstanding Undergraduate Thesis, SUSTech *2025*
- **National Scholarship, Ministry of Education of China (Top 1%)** *2024*
- First-class Scholarship, SUSTech *2023, 2024*
- Top 10 Volunteers, SUSTech *2022*

Research Experience

Conductive-Silicon-Packaged Piezoelectric Resonators

*Advised by Prof. **Ruochen Lu**, UT*

Apr. 2026 – Present

- Designed a conductive-silicon package for piezoelectric resonators, demonstrated with LiNbO₃ in a Si–LN–Si stack, to improve power handling, avoid electrode breakdown and enhance heat dissipation.
- Demonstrated capacitive excitation of LN through Si–LN air gaps using conductive Si electrodes, retaining 95% and 99% of the bare-LN Q and k^2 , respectively.
- Fabricated Si–LN–Si resonator prototypes using multiple bonding methods (Au–Au, BCB, photoresist, and Crystalbond) and evaluated their effects on resonator performance.
- Evaluated packaging effects on power handling and thermal performance under high-power excitation.

Vertically Stacked LiNbO₃ Isolation Transformers

Advised by Prof. **Ruochen Lu, UT**

Mar. 2026 – Present

- Designed a vertically stacked LN–sapphire–LN isolation transformer for mechanically coupled power transfer across an electrically insulating layer, targeting high power handling and high breakdown voltage.
- Simulated mode shapes and admittance spectra of 36°, 128°, and 163° Y-cut LN in COMSOL; optimized layer thicknesses for half-wavelength resonance to improve acoustic transmission.
- Fabricated two-port transformer prototypes and designed test PCBs with corner-fixed mounting and wire-bonded electrical interconnects.
- Currently characterizing resonance frequencies, Q , and k^2 using VNA measurements; evaluating nonlinearity, power handling, temperature coefficient of frequency (TCF), and power-transfer efficiency.

Modeling and Control of Multiphase Buck Converters

Advised by Prof. **Yicheng Zhu, UT**

Feb. 2026 – Jun. 2026

- Developed state-space-averaged and small-signal models of multiphase buck converters; analyzed converter dynamics and loop-gain characteristics for feedback-controller design.
- Implemented voltage-mode, current-mode, COT and V^2 control in PLECS; evaluated closed-loop stability through frequency-domain analysis and dynamic regulation through time-domain simulations.
- Compared transient responses across different control strategies, quantifying trade-offs in voltage deviation, settling time, and implementation complexity.

MEMS Gas Sensor Arrays with Wafer-Level Material Patterning

Advised by Prof. **Fei Wang, SUSTech**

Jun. 2023 – Apr. 2025

- Optimized a backside-etched microheater structure, reducing heating power consumption by 25% compared with the frontside-etched design.
- Developed a wafer-level multi-step photolithography process for patterning multiple gas-sensing materials on MEMS sensor arrays, and fabricated devices through lithography, deposition and etching.
- Applied support vector machine (SVM) models to sensor-array responses, achieving 100% gas classification accuracy and $R^2 \geq 0.98$ for concentration prediction on a test set.

Course Projects

24 V to 48 V Synchronous Boost Converter Prototype

Jan. 2026 – Jun. 2026

Instructor: Prof. **Yicheng Zhu**

Course: *Power Electronic Devices and Systems*

- Designed and laid out a 250 W, 24 V to 48 V synchronous boost converter on a four-layer PCB in **KiCad**, minimizing high- di/dt commutation and gate-drive loop areas to reduce parasitic inductance.
- Designed and built the power inductor and programmed a TI C2000 MCU for complementary PWM with dead time and synchronized ADC sensing of input/output voltages and inductor current.
- Assembled and tested the converter under load; characterized switching waveforms, voltage/current ripple, and load-dependent efficiency. Achieved a peak conversion efficiency of 95.7%.

Synchronous Buck Converter Design with V^2 Peak Control

Sep. 2025 – Nov. 2025

Instructor: Prof. **Alex Q. Huang**

Course: *PMIC*

- Designed and simulated a synchronous buck converter with V^2 peak control in **Cadence Virtuoso**, operating in PWM mode at high load and PFM mode at light load for enhanced efficiency.
- Integrated a complete control system, incorporating ramp generator, comparator, compensation network, gate drivers with dead-time control, and OCP and OVP protection.
- Optimized transistor sizing and passive component dimensions to minimize the overall design footprint.

Technical skills & Professional services

- **Design & Simulation:** COMSOL Multiphysics, KiCad, PLECS, LTspice, Cadence Virtuoso, Silvaco TCAD.
- **Microfabrication:** Photolithography, Thin-film Deposition (Sputtering, Evaporation, ALD), Etching (ICP, DRIE, KOH/BOE), Wafer Bonding/Dicing.
- **Programming:** C++, MATLAB, Python, Verilog, Java.
- **Reviewer:** Micro and Nano Engineering.